

Harish Kumar A

Robotics Software Engineer | Autonomous Systems | ROS 2 | Embedded Robotics

+91 9360742586 | hku.kumar.dev@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Robotics Software Engineer with 1 year of hands-on industry experience developing, integrating, testing, and debugging autonomous AGV and AMR systems across navigation, localization, perception, motion control, and embedded integration. B.Tech Artificial Intelligence and Data Science graduate with experience building and validating software on physical robotic platforms.

EXPERIENCE

Software Engineer

Mar. 2026 – Aug. 2026

Lambda Robotics

Kovilpatti, Tamil Nadu

- Developed and validated navigation, localization, motion-control, and embedded-communication functions for autonomous AGV and AMR platforms using Python, C++, Linux, and ROS 2.
- Integrated LiDAR, cameras, IMUs, encoders, actuators, motor drivers, Raspberry Pi, and ESP32 controllers across physical robotic platforms for perception, localization, control, diagnostics, and field testing.
- Developed modular PyQt6 HMI components for task assignment, mission monitoring, battery and Wi-Fi status, robot interaction, and bidirectional Python-JavaScript communication using Qt WebChannel.
- Diagnosed and resolved system-level failures involving serial communication, sensor accuracy, IMU heading drift, motor direction, camera connectivity, networking, wiring, and power sequencing through structured root-cause analysis.

Software Engineer Intern

Sep. 2025 – Feb. 2026

Lambda Robotics

Kovilpatti, Tamil Nadu

- Contributed to robotics research and prototype development involving autonomous navigation, computer vision, embedded systems, sensors, motors, electronics integration, mechanical design, and early AGV development.
- Developed and tested a Wi-Fi-controlled obstacle-avoiding robot and supported robotics prototyping, system integration, and physical robot validation.

PROJECTS

Vision & IMU-Based Autonomous Guided Vehicle | *Python, C++, OpenCV, AprilTags, ESP32*

Mar. 2026 – Aug. 2026

- Built and field-tested an indoor differential-drive AGV using AprilTag-based localization, IMU heading feedback, encoder data, A* path planning, lateral course correction, and closed-loop turning.
- Implemented differential-drive motion calculations, jerk-limited acceleration, predictive braking, and Raspberry Pi-ESP32 communication for velocity commands, localization correction, and motor control.
- Developed runtime diagnostic telemetry for heading, gyro rate, lateral error, wheel velocity, pulse rate, acceleration, and travelled distance; achieved stable operation at approximately 0.245 m/s and 40 wheel RPM.

Autonomous Mobile Service Robot | *ROS 2, SLAM, LiDAR, Depth Camera, YOLO*

May 2026

- Contributed to an autonomous indoor service robot by integrating LiDAR and depth-camera inputs with ROS 2 SLAM-based mapping and localization for navigation in a customer-facing cafe environment.
- Applied YOLO-based person and obstacle detection to support environment-aware speed reduction and stopping behavior around customers and surrounding objects.

AGV Vehicle Control Unit & Power Management | *ESP32, Raspberry Pi, GPIO, Linux, systemd*

Mar. 2026 – May 2026

- Designed an ESP32-based Vehicle Control Unit and Raspberry Pi architecture for coordinated AGV control, embedded communication, startup sequencing, and power management.
- Implemented relay-based power sequencing, power-hold logic, graceful Linux shutdown, and GPIO handshaking for boot readiness, shutdown requests, shutdown completion, and controller supervision.
- Configured systemd services to automatically initialize robot software during startup and coordinate application readiness with the embedded power-control workflow.

Six-Axis Robotic Arm with Cycloidal Gear Reduction | *Kinematics, Mechanical Design, FreeCAD*

Feb. 2026

- Contributed to the conceptual development of a six-axis robotic arm for an automated dishwashing application, evaluating compact high-reduction transmission mechanisms for robotic joints.
- Designed and evaluated cycloidal gear components and studied forward/inverse kinematics, joint coordinate systems, workspace, singularities, joint motion, and critical load points affecting manipulator stability.

TECHNICAL SKILLS

Programming: Python, C++, C, Java

Robotics & Autonomy: ROS 2, AGV/AMR, Autonomous Navigation, Localization, SLAM, A* Path Planning, Motion Control, Differential-Drive Kinematics, Sensor Fusion

Perception: OpenCV, YOLO, AprilTags, LiDAR, Depth Cameras, IMU, Encoders, Real-Time Image Processing

Embedded & Hardware: ESP32, Raspberry Pi, Arduino, Teensy, UART, I2C, GPIO, Relays, Motor Drivers, Stepper Motors, Serial Communication

Systems & Tools: Linux, systemd, PyQt6, Qt WebEngine, Qt WebChannel, Git, GitHub, Docker, Webots, FreeCAD, SSH, VNC

EDUCATION

Dr. Mahalingam College of Engineering and Technology

Pollachi, Tamil Nadu

B.Tech in Artificial Intelligence and Data Science; CGPA: 8.14/10

2022 – 2026